



Engineering Electromagnetics

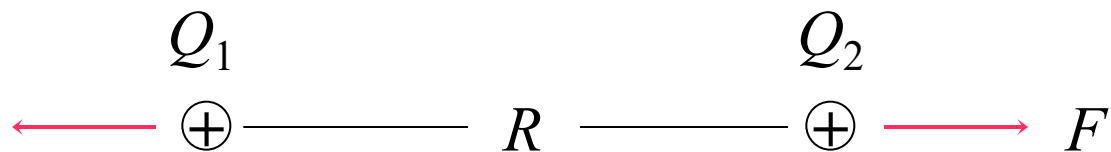
工程电磁场

Youwei Jia 嘉有为

2: Coulomb's Law 库伦定律



- Coulomb's electrostatic force law 库伦静电力定律
- Charles Augustin de Coulomb (1736 – 1806)
French physicist, French Army Engineers
- Coulomb's torsion balance 库伦扭秤



Force of repulsion 排斥力, F , occurs when charges have the same sign.

Charges attract each other when of opposite sign.

$$F = k \frac{Q_1 Q_2}{R^2} \quad \text{where} \quad k = \frac{1}{4\pi\epsilon_0}$$



Note: Please compare the Coulomb's electrostatic force law with the Newton's law of universal gravitation.

2: Coulomb's Law 库伦定律



- Free Space Permittivity 自由空间的介电常数

$$F = k \frac{Q_1 Q_2}{R^2} \quad \text{where} \quad k = \frac{1}{4\pi \epsilon_0}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \doteq \frac{1}{36\pi} 10^{-9} \text{ F/m}$$

F: Farad 法拉/法

with which the Coulomb's force becomes:

$$F = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2}$$

- Unit of Charge 电荷量单位: Coulomb, C 库伦, 库
- Elementary charge 基本电荷量 (Charge of Electron or Proton)
 $= 1.60217657 \times 10^{-19} \text{ Coulombs}$

2: Coulomb's Law 库伦定律



- Coulomb's law in vector form 库伦定律的矢量形式

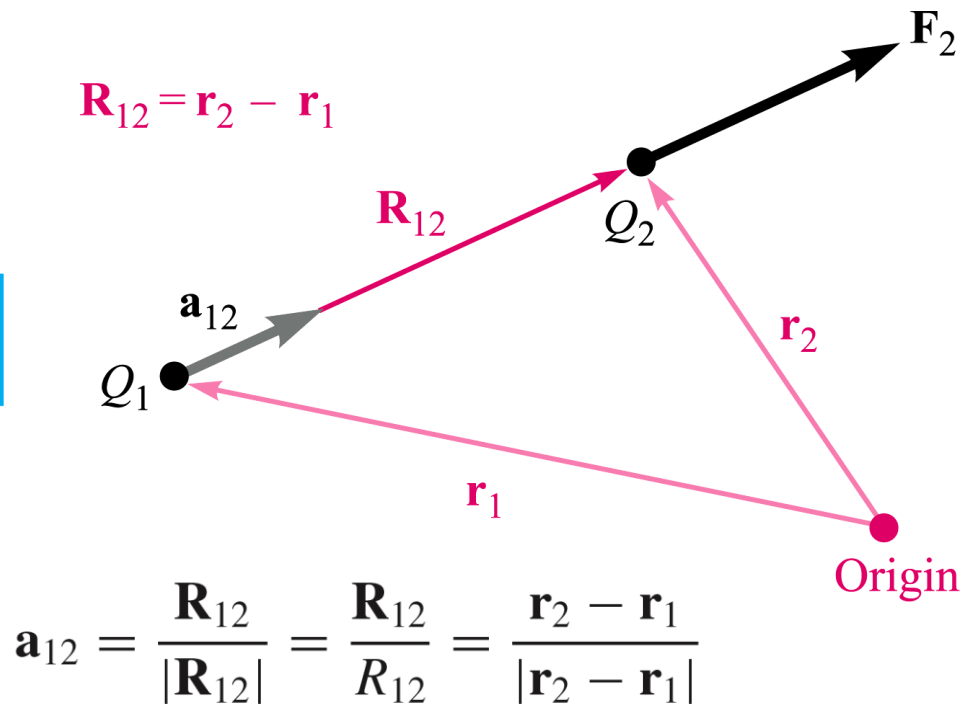
$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2}$$



$$\mathbf{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12}$$

Q1. How to describe $\mathbf{F}_1$?

$$\mathbf{F}_1 = -\mathbf{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{21} = -\frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12}$$



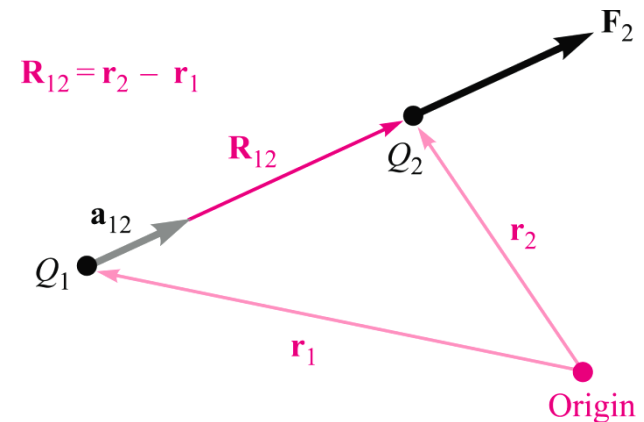
2: Coulomb's Law 库伦定律



- Example

$$\mathbf{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12}$$

$$\mathbf{a}_{12} = \frac{\mathbf{R}_{12}}{|\mathbf{R}_{12}|} = \frac{\mathbf{R}_{12}}{R_{12}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|}$$



We illustrate the use of the vector form of Coulomb's law by locating a charge of $Q_1 = 3 \times 10^{-4} \text{ C}$ at $M(1, 2, 3)$ and a charge of $Q_2 = -10^{-4} \text{ C}$ at $N(2, 0, 5)$ in a vacuum. We desire the force exerted on Q_2 by Q_1 .

Solution. We use (3) and (4) to obtain the vector force. The vector $\mathbf{R}_{12}$ is

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1 = (2 - 1)\mathbf{a}_x + (0 - 2)\mathbf{a}_y + (5 - 3)\mathbf{a}_z = \mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z$$

leading to $|\mathbf{R}_{12}| = 3$, and the unit vector, $\mathbf{a}_{12} = \frac{1}{3}(\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z)$. Thus,

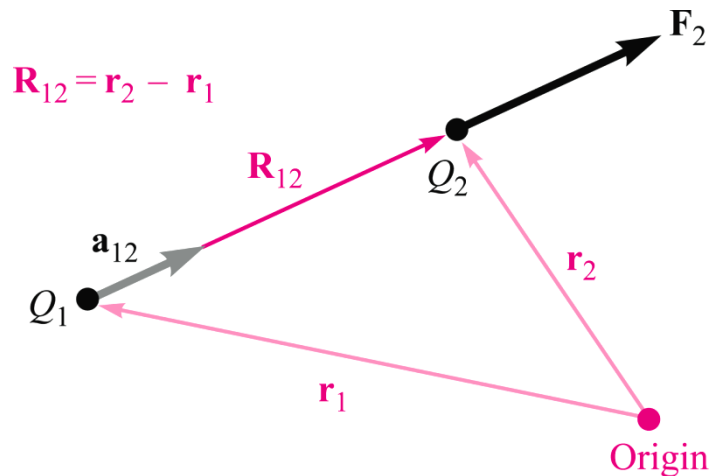
$$\begin{aligned} \mathbf{F}_2 &= \frac{3 \times 10^{-4}(-10^{-4})}{4\pi(1/36\pi)10^{-9} \times 3^2} \left(\frac{\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z}{3} \right) \\ &= -30 \left(\frac{\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z}{3} \right) \text{ N} \end{aligned}$$

3: Electric Field Intensity 电场强度



- Unit test charge 单位试验电荷

$$\mathbf{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12}$$



$$\mathbf{a}_{12} = \frac{\mathbf{R}_{12}}{|\mathbf{R}_{12}|} = \frac{\mathbf{R}_{12}}{R_{12}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|}$$

Consider the force acting on a **test charge**, Q_t , arising from charge Q_1 :

$$\mathbf{F}_t = \frac{Q_1 Q_t}{4\pi\epsilon_0 R_{1t}^2} \mathbf{a}_{1t}$$

where $\mathbf{a}_{1t}$ is the unit vector directed from Q_1 to Q_t

The **electric field intensity** is defined as the force per unit test charge, or

$$\mathbf{E}_1 = \frac{\mathbf{F}_t}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 R_{1t}^2} \mathbf{a}_{1t} \quad \text{N/C}$$

3: Electric Field Intensity 电场强度



- Electric field intensity

$$\mathbf{E}_1 = \frac{\mathbf{F}_t}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 R_{1t}^2} \mathbf{a}_{1t} \quad \text{N/C}$$



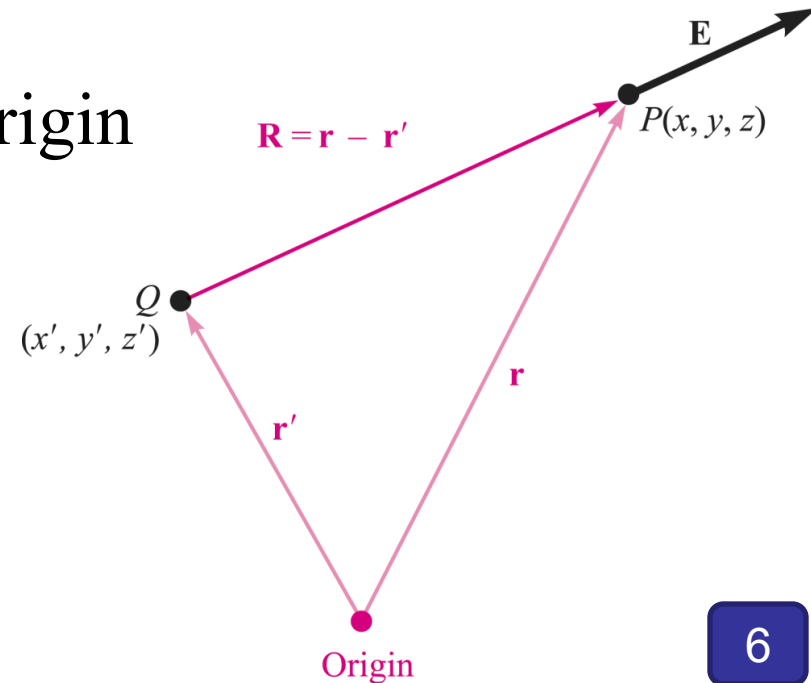
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

A more convenient unit for electric field is **V/m**. V: Volt 伏特

Q2. What is $\mathbf{a}_R$?

- Electric field of a charge off-origin

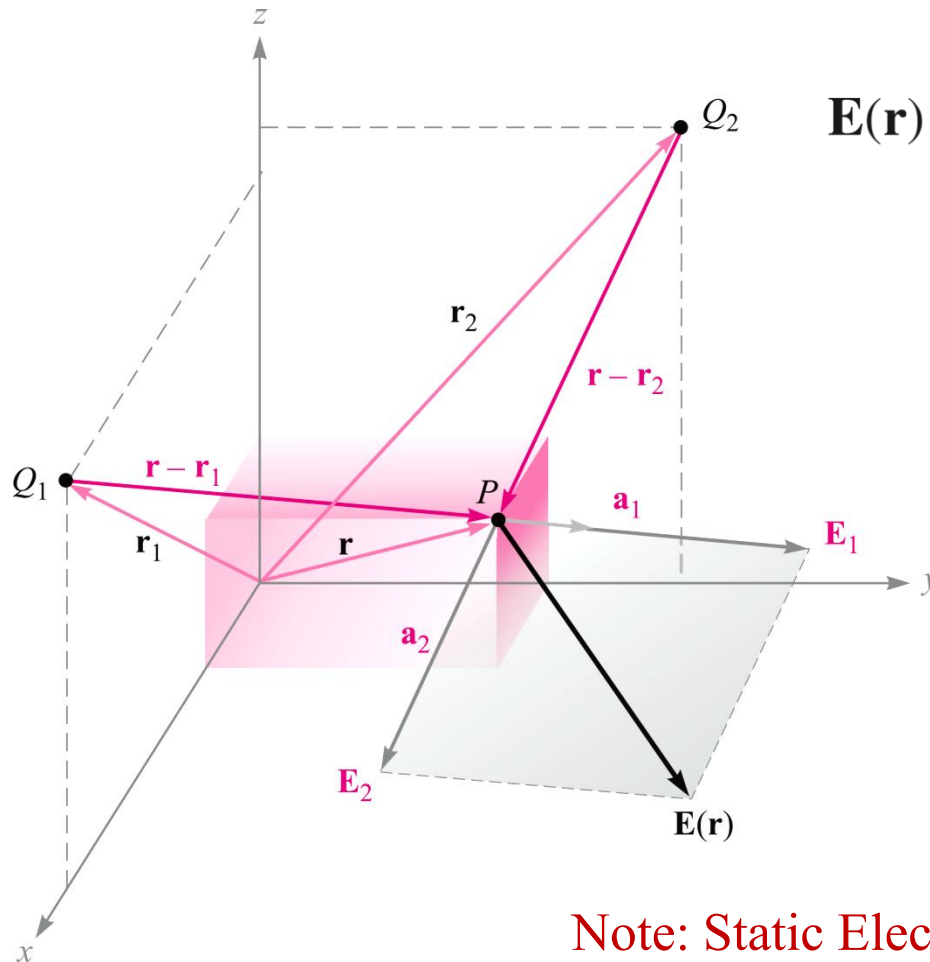
$$\begin{aligned} \mathbf{E}(\mathbf{r}) &= \frac{Q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3} \\ &= \frac{Q[(x - x')\mathbf{a}_x + (y - y')\mathbf{a}_y + (z - z')\mathbf{a}_z]}{4\pi\epsilon_0 [(x - x')^2 + (y - y')^2 + (z - z')^2]^{3/2}} \end{aligned}$$



3: Electric Field Intensity 电场强度



- Superposition of fields from two point charges 电场叠加



$$\mathbf{E}(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_1|^2}\mathbf{a}_1 + \frac{Q_2}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_2|^2}\mathbf{a}_2$$

For n charges:

$$\mathbf{E}(\mathbf{r}) = \sum_{m=1}^n \frac{Q_m}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_m|^2}\mathbf{a}_m$$

Note: Static Electric Field in free space is linear!

3: Electric Field Intensity 电场强度



- Example

Find $\mathbf{E}$ at P , using $\mathbf{E}(\mathbf{r}) = \sum_{m=1}^n \frac{Q_m}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_m|^2} \mathbf{a}_m$

$$Q_m = 3nC$$

First, find the vectors:

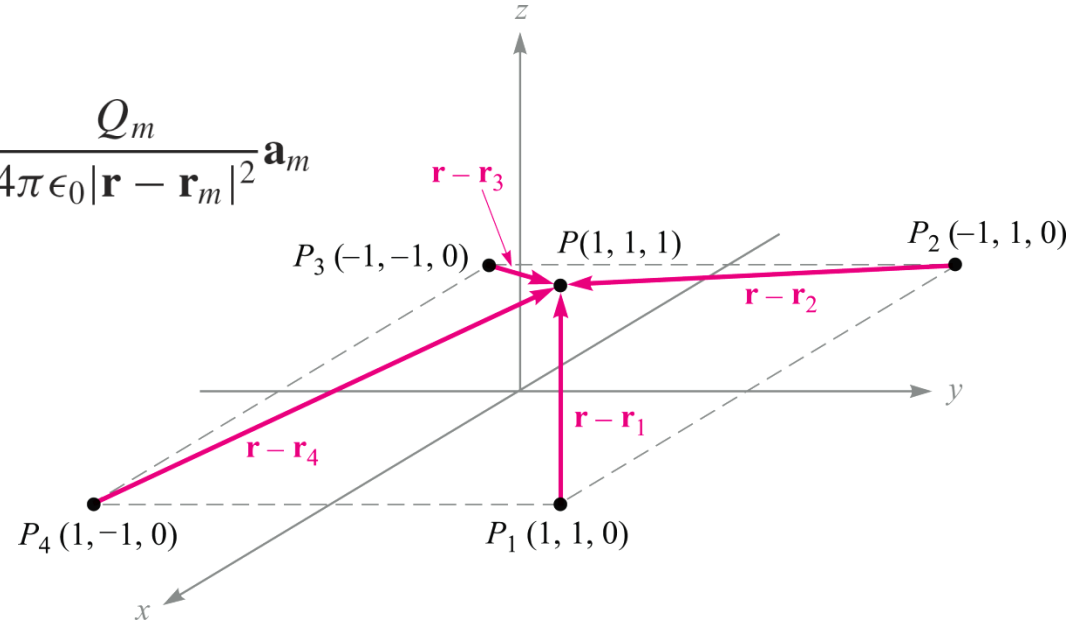
$$\mathbf{r} = \mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z$$

$$\mathbf{r}_1 = \mathbf{a}_x + \mathbf{a}_y$$

$$\mathbf{r}_2 = -\mathbf{a}_x + \mathbf{a}_y$$

$$\mathbf{r}_3 = -\mathbf{a}_x - \mathbf{a}_y$$

$$\mathbf{r}_4 = \mathbf{a}_x - \mathbf{a}_y$$



Then:

$$|\mathbf{r} - \mathbf{r}_1| = 1$$

$$|\mathbf{r} - \mathbf{r}_2| = \sqrt{5}$$

$$|\mathbf{r} - \mathbf{r}_3| = 3$$

$$|\mathbf{r} - \mathbf{r}_4| = \sqrt{5}$$

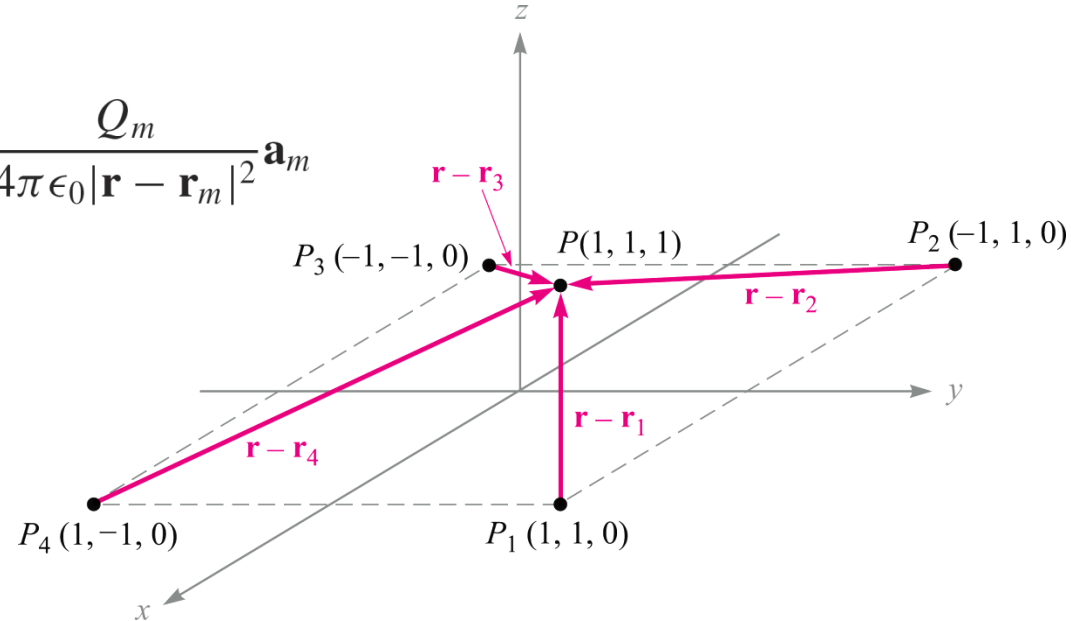
3: Electric Field Intensity 电场强度



- Example (Cont.'d)

Find $\mathbf{E}$ at P , using $\mathbf{E}(\mathbf{r}) = \sum_{m=1}^n \frac{Q_m}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_m|^2} \mathbf{a}_m$

where $\mathbf{a}_m = \frac{\mathbf{r} - \mathbf{r}_m}{|\mathbf{r} - \mathbf{r}_m|}$



$$Q/4\pi\epsilon_0 = 3 \times 10^{-9} / (4\pi \times 8.854 \times 10^{-12}) = 26.96 \text{ V} \cdot \text{m}$$

so that:

$$\begin{aligned} \mathbf{E} &= 26.96 \left[\frac{\mathbf{a}_z}{1} \frac{1}{1^2} + \frac{2\mathbf{a}_x + \mathbf{a}_z}{\sqrt{5}} \frac{1}{(\sqrt{5})^2} + \frac{2\mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z}{3} \frac{1}{3^2} + \frac{2\mathbf{a}_y + \mathbf{a}_z}{\sqrt{5}} \frac{1}{(\sqrt{5})^2} \right] \\ &= \underline{6.82\mathbf{a}_x + 6.82\mathbf{a}_y + 32.8\mathbf{a}_z \text{ V/m}} \end{aligned}$$

3: Electric Field Intensity 电场强度



- Electric field arising from a continuous volume charge distribution 连续分布体电荷产生的电场

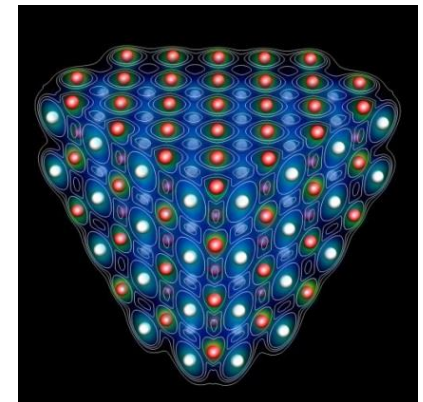
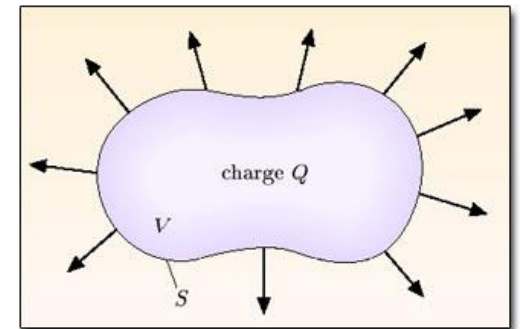
The incremental contribution to the electric field intensity at $\mathbf{r}$ produced by an incremental charge ΔQ at $\mathbf{r}'$ is

$$\Delta \mathbf{E}(\mathbf{r}) = \frac{\Delta Q}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$

We denote **volume charge density** 体电荷密度 by ρ_v , having the units of **coulombs per cubic meter** (C/m^3):

$$\Delta Q = \rho_v \Delta v \quad \rho_v = \lim_{\Delta v \rightarrow 0} \frac{\Delta Q}{\Delta v}$$

$$\Delta \mathbf{E}(\mathbf{r}) = \frac{\rho_v \Delta v}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$



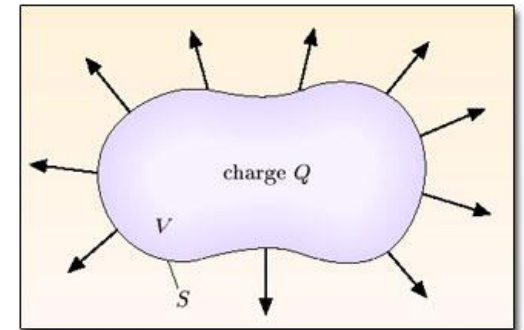
3: Electric Field Intensity 电场强度



- Electric field arising from a continuous volume charge distribution 连续分布体电荷产生的电场

Next, sum all contributions throughout a volume and take the limit as Δv approaches zero, to obtain the integral:

$$\Delta \mathbf{E}(\mathbf{r}) = \frac{\rho_v \Delta v}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$



$$\mathbf{E}(\mathbf{r}) = \int_{\text{vol}} \frac{\rho_v(\mathbf{r}') dv'}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$

$$Q = \int_{\text{vol}} \rho_v dv$$

Note: The vector $\mathbf{r}$ extends from the origin to the field point where $\mathbf{E}$ is being determined, whereas the vector $\mathbf{r}'$ extends from the origin to the source point where $\rho_v(\mathbf{r}')dv'$ is located.

3: Electric Field Intensity 电场强度



- Electric field arising from a continuous volume charge distribution 连续分布体电荷产生的电场

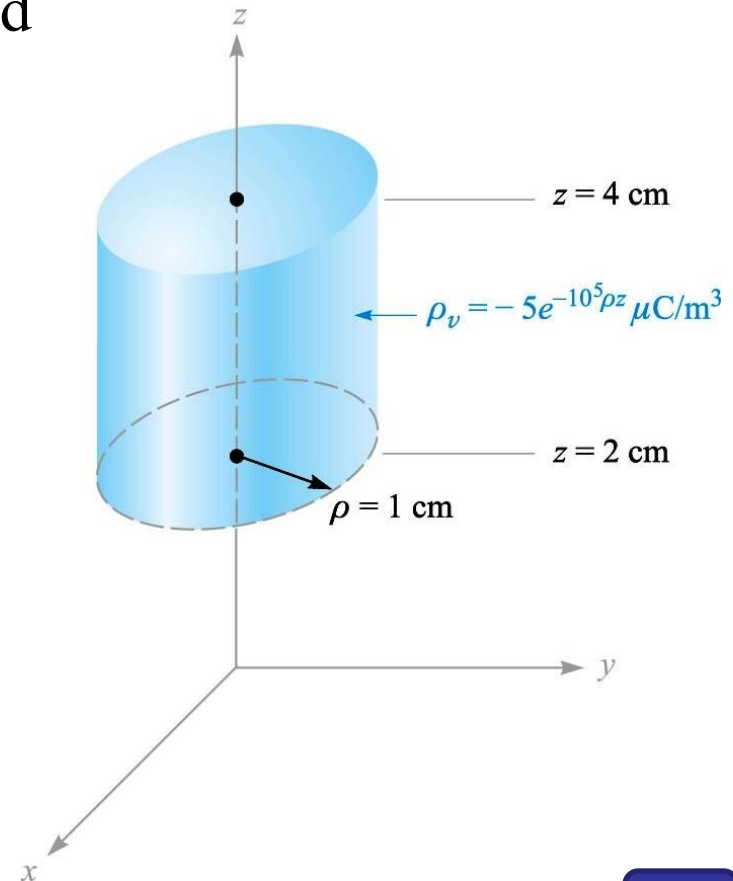
Example: Find the total charge contained in a 2-cm length of the electron beam

$$\rho_v = -5 \times 10^{-6} e^{-10^5 \rho z} \text{ C/m}^3$$

$$Q = \int_{0.02}^{0.04} \int_0^{2\pi} \int_0^{0.01} -5 \times 10^{-6} e^{-10^5 \rho z} \rho \, d\rho \, d\phi \, dz$$

$$Q = \int_{0.02}^{0.04} \int_0^{0.01} -10^{-5} \pi e^{-10^5 \rho z} \rho \, d\rho \, dz$$

$$\begin{aligned} Q &= \int_0^{0.01} \left(\frac{-10^{-5} \pi}{-10^5 \rho} e^{-10^5 \rho z} \rho \, d\rho \right)_{z=0.02}^{z=0.04} \\ &= \int_0^{0.01} -10^{-5} \pi (e^{-2000\rho} - e^{-4000\rho}) d\rho \end{aligned}$$



3: Electric Field Intensity 电场强度



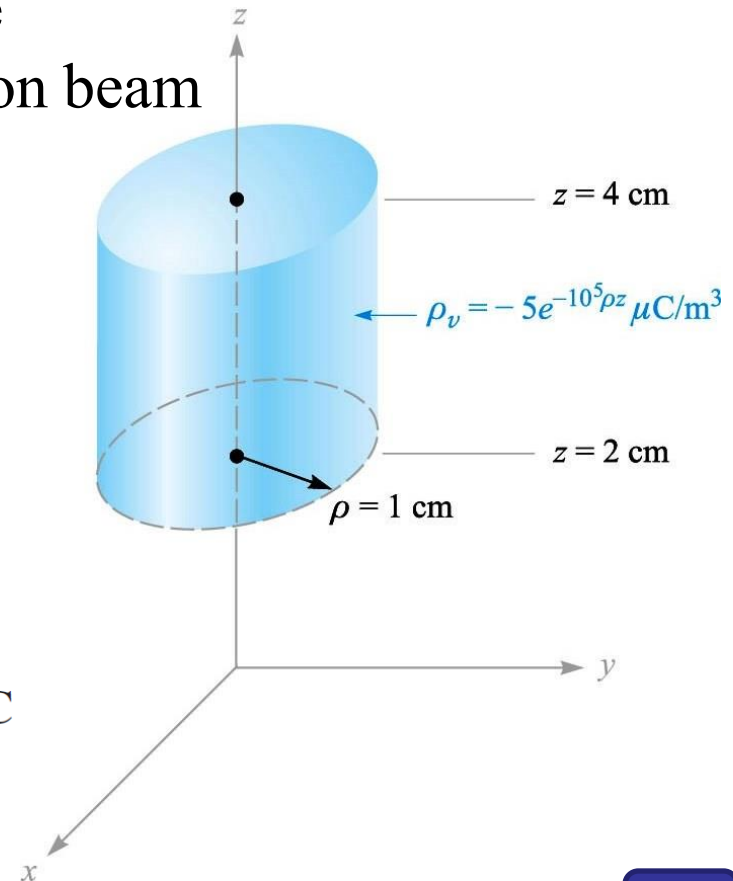
- Electric field arising from a continuous volume charge distribution 连续分布体电荷产生的电场

Example (Cont'd): Find the total charge contained in a 2-cm length of the electron beam

$$Q = \int_0^{0.01} \left(\frac{-10^{-5}\pi}{-10^5\rho} e^{-10^5\rho z} \rho d\rho \right)_{z=0.02}^{z=0.04}$$
$$= \int_0^{0.01} -10^{-5}\pi (e^{-2000\rho} - e^{-4000\rho}) d\rho$$

$$Q = -10^{-10}\pi \left(\frac{e^{-2000\rho}}{-2000} - \frac{e^{-4000\rho}}{-4000} \right)_0^{0.01}$$

$$Q = -10^{-10}\pi \left(\frac{1}{2000} - \frac{1}{4000} \right) = \frac{-\pi}{40} = 0.0785 \text{ pC}$$



Note: pico-coulombs (pc皮库), 10^{-12}C

3: Electric Field Intensity 电场强度



- Electric field of a line charge 线电荷产生的电场

Line charge of constant line charge density 线电荷密度 ρ_L C/m lies along the entire z axis. (infinite line of charge)

At point P , the electric field arising from charge dQ on the z axis is:

$$d\mathbf{E} = \frac{\rho_L dz' (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

where $\mathbf{r} = \rho \mathbf{a}_\rho = \rho \mathbf{a}_x$

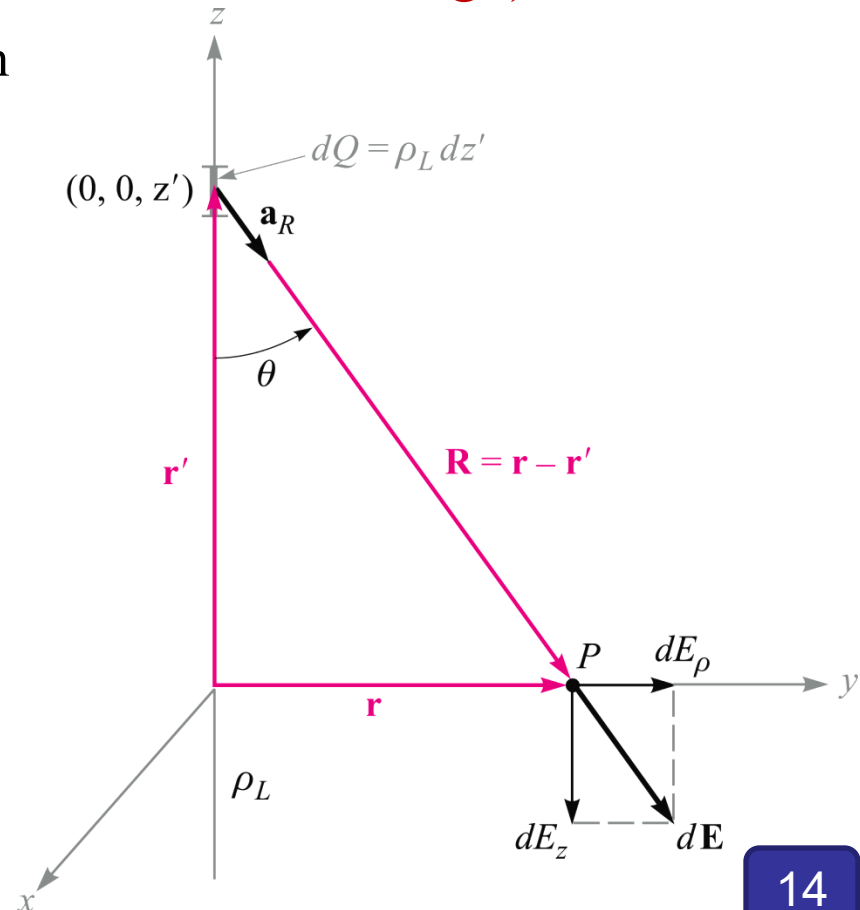
and $\mathbf{r}' = z' \mathbf{a}_z$

so that $\mathbf{r} - \mathbf{r}' = \rho \mathbf{a}_\rho - z' \mathbf{a}_z$

Therefore

$$d\mathbf{E} = \frac{\rho_L dz' (\rho \mathbf{a}_\rho - z' \mathbf{a}_z)}{4\pi\epsilon_0 (\rho^2 + z'^2)^{3/2}}$$

Note: Cylindrical coordinate is adopted herein.



3: Electric Field Intensity 电场强度



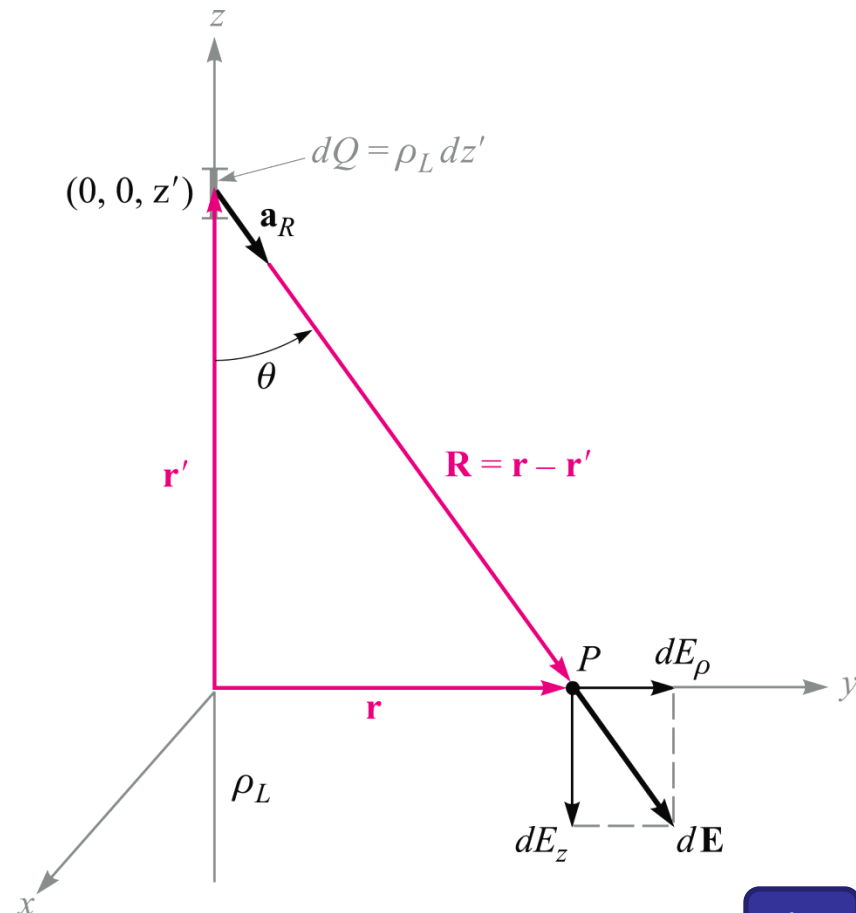
• Electric field of a line charge 线电荷产生的电场

We have:
$$d\mathbf{E} = \frac{\rho_L dz'(\rho \mathbf{a}_\rho - z' \mathbf{a}_z)}{4\pi\epsilon_0(\rho^2 + z'^2)^{3/2}}$$

By symmetry, only a radial component is present:

$$dE_\rho = \frac{\rho_L \rho dz'}{4\pi\epsilon_0(\rho^2 + z'^2)^{3/2}}$$

$$\begin{aligned} E_\rho &= \int_{-\infty}^{\infty} \frac{\rho_L \rho dz'}{4\pi\epsilon_0(\rho^2 + z'^2)^{3/2}} \\ &= \frac{\rho_L}{4\pi\epsilon_0} \rho \left(\frac{1}{\rho^2} \frac{z'}{\sqrt{\rho^2 + z'^2}} \right)_{-\infty}^{\infty} \end{aligned}$$



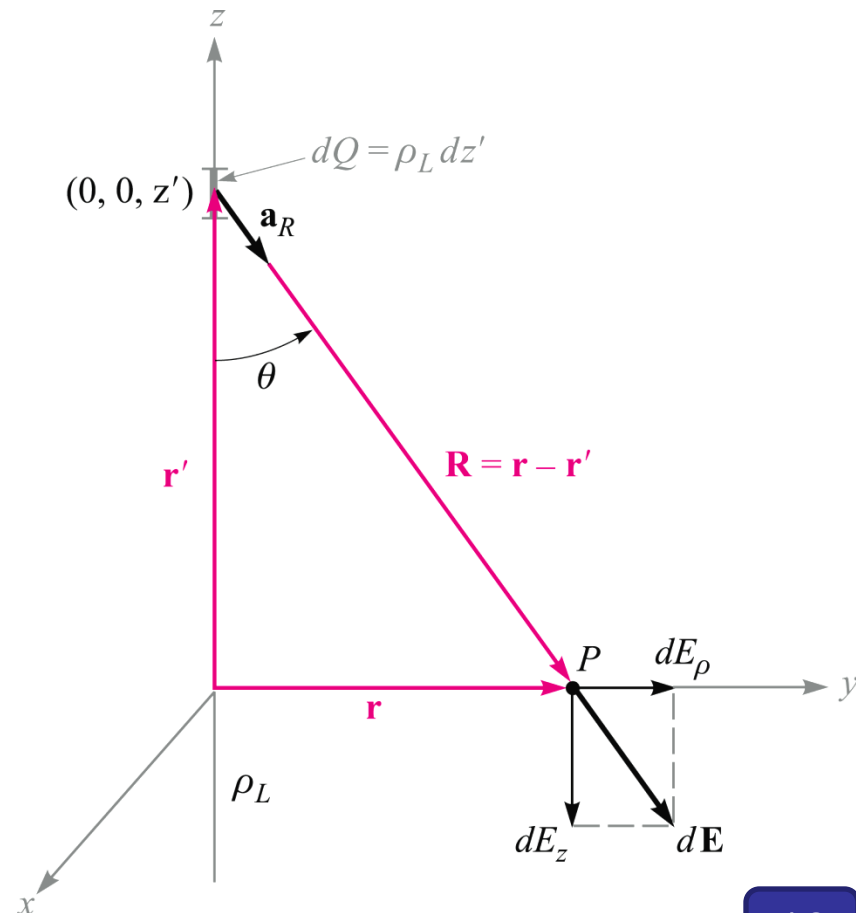
3: Electric Field Intensity 电场强度



- Electric field of a line charge 线电荷产生的电场

$$E_{\rho} = \frac{\rho_L}{4\pi\epsilon_0} \rho \left(\frac{1}{\rho^2} \frac{z'}{\sqrt{\rho^2 + z'^2}} \right)_{-\infty}^{\infty}$$
$$= \frac{\rho_L}{2\pi\epsilon_0\rho}$$

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_{\rho}$$



Q3. How to describe in Rectangular coordinates?

Q4. How to choose coordinate system?

Q5. How to memorize this result?

3: Electric Field Intensity 电场强度



• Electric field of a line charge 线电荷产生的电场

Example: Off-Axis Line Charge

With the line displaced to (6,8), the field becomes:

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\sqrt{(x-6)^2 + (y-8)^2}} \mathbf{a}_R$$

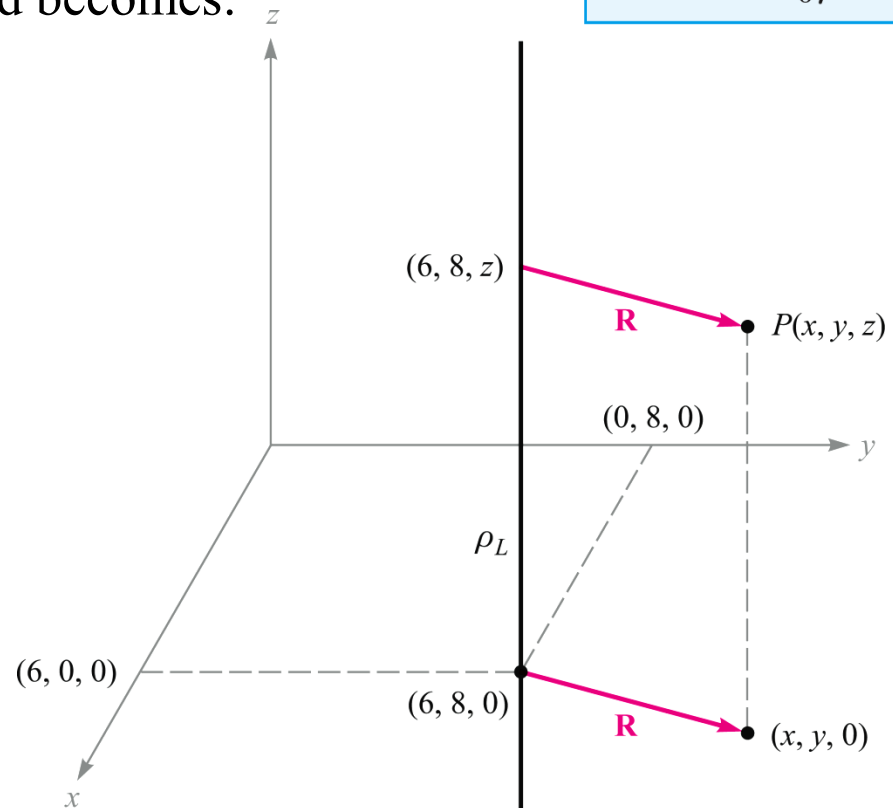
where

$$\mathbf{a}_R = \frac{\mathbf{R}}{|\mathbf{R}|} = \frac{(x-6)\mathbf{a}_x + (y-8)\mathbf{a}_y}{\sqrt{(x-6)^2 + (y-8)^2}}$$

Finally:

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0} \frac{(x-6)\mathbf{a}_x + (y-8)\mathbf{a}_y}{(x-6)^2 + (y-8)^2}$$

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$



3: Electric Field Intensity 电场强度

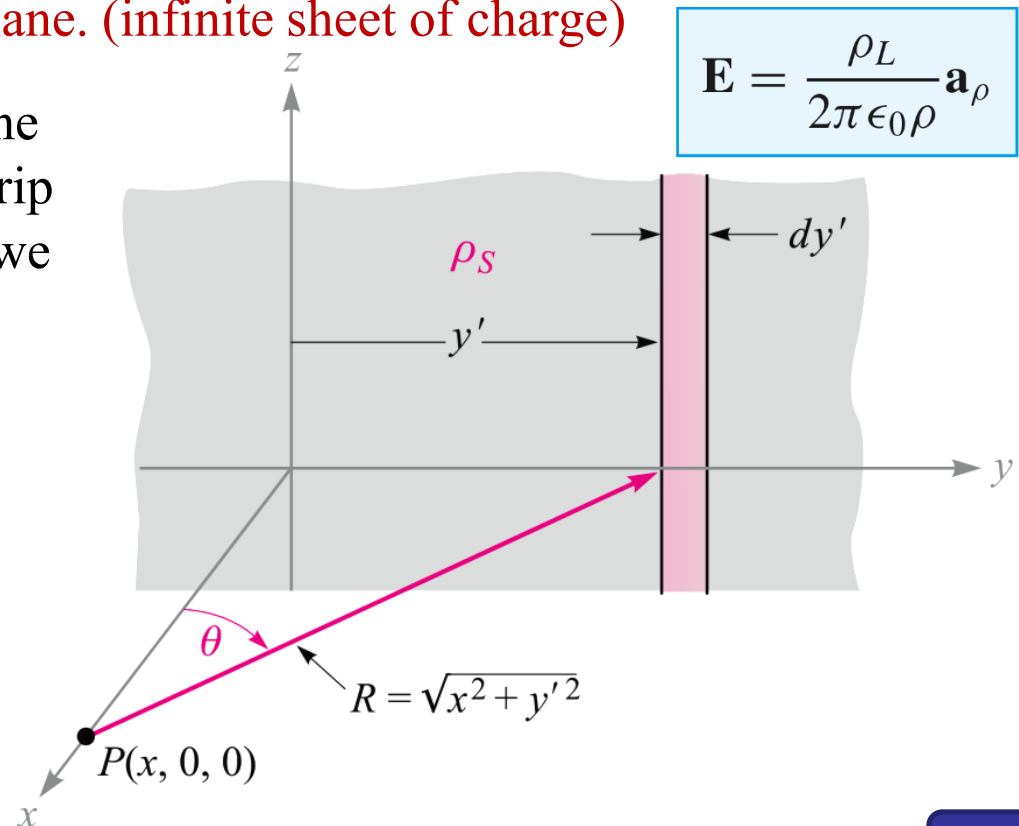


- Electric field of a sheet of charge 面电荷产生的电场
Surface charge density 面电荷密度 $\rho_s \text{ C/m}^2$

Consider the electric field produced by the uniform surface charge of density ρ_s covers the entire y - z plane. (infinite sheet of charge)

We begin by writing down the line charge field on the x axis for a strip of differential width dy' , where we consider the x component of that field:

$$\begin{aligned} dE_x &= \frac{\rho_s dy'}{2\pi\epsilon_0\sqrt{x^2 + y'^2}} \cos\theta \\ &= \frac{\rho_s}{2\pi\epsilon_0} \frac{xdy'}{x^2 + y'^2} \end{aligned}$$



3: Electric Field Intensity 电场强度



- Electric field of a sheet of charge 面电荷产生的电场
Surface charge density 面电荷密度 $\rho_S \text{ C/m}^2$

$$E_x = \frac{\rho_S}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{x dy'}{x^2 + y'^2} = \frac{\rho_S}{2\pi\epsilon_0} \left[\tan^{-1} \frac{y'}{x} \right]_{-\infty}^{\infty} = \frac{\rho_S}{2\epsilon_0}$$

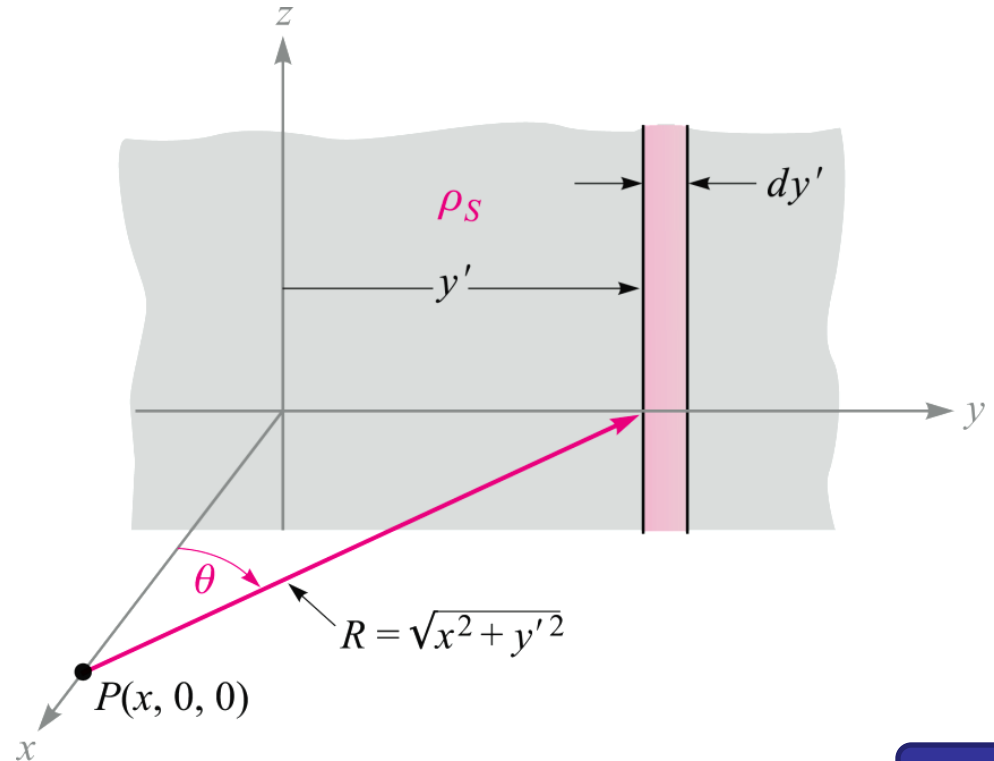
The total field on the x axis is:

$$\mathbf{E} = \frac{\rho_S}{2\epsilon_0} \mathbf{a}_N$$

Q5. How to memorize this result?

Q6. Comparing P15 & P18, what do you think?

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0 \rho} \mathbf{a}_\rho$$



3: Electric Field Intensity 电场强度

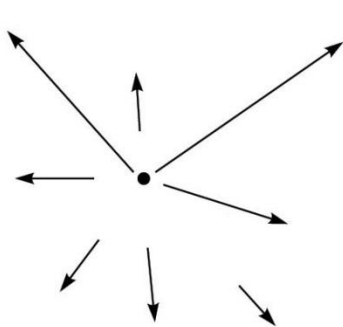


- Streamlines and sketches of fields 电力线和电场分布图

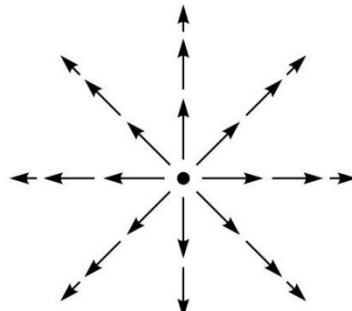
How to visualize a electric field?

Consider the electric field caused by line charge

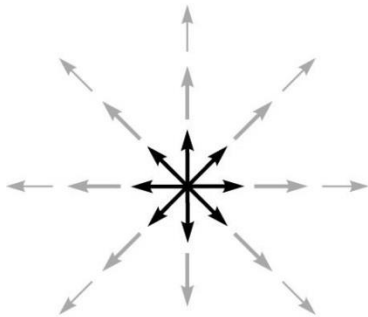
$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$



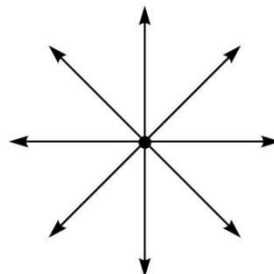
(a)



(b)



(c)



(d)

Let us be content to show only the *direction* of $\mathbf{E}$ by drawing **continuous lines**, which are everywhere **tangent to $\mathbf{E}$** , from the charge.

These lines are usually called **streamlines**, although other terms such as **flux lines** 通量线 and **direction lines** 方向线 are also used.

Q6. Is it real that the magnitude of field can not be indicated with streamlines?

3: Electric Field Intensity 电场强度

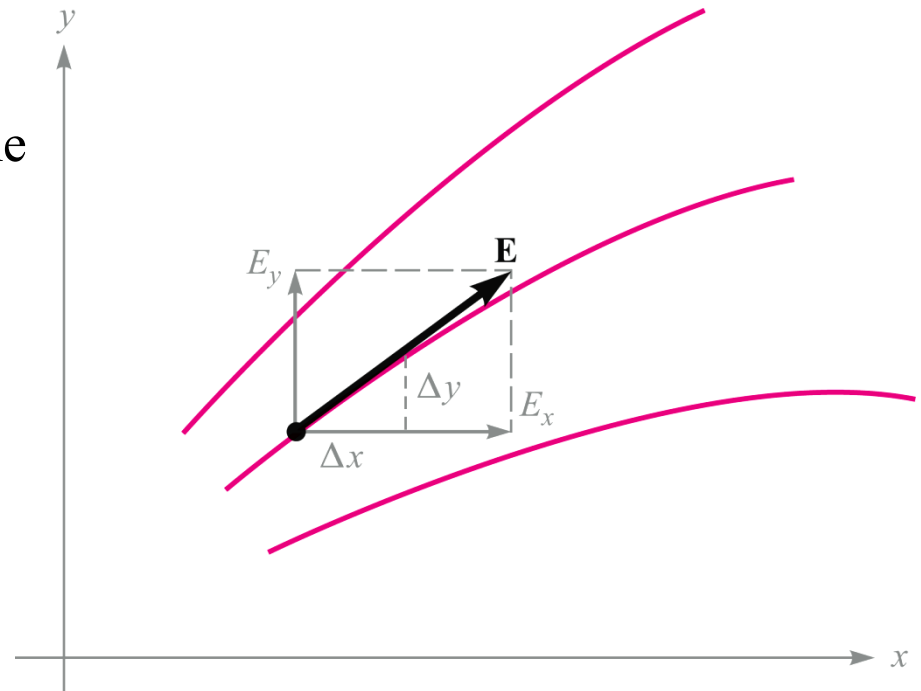


- Streamlines and sketches of fields 电力线和电场分布图

Methodology of Streamline Construction 电力线的构建方法

The ratio of the y and x field components gives the slope of the field plot in the x - y plane

$$\frac{E_y}{E_x} = \frac{dy}{dx}$$



3: Electric Field Intensity 电场强度



- Streamlines and sketches of fields 电力线和电场分布图

Methodology of Streamline Construction 电力线的构建方法

Begin with the **normalized** line charge field in cylindrical coordinates:

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho \quad \rho_L = 2\pi\epsilon_0 \quad \mathbf{E} = \frac{1}{\rho}\mathbf{a}_\rho$$

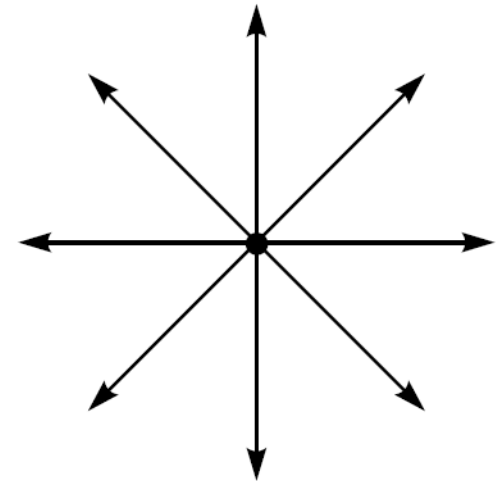
Convert to rectangular components:

$$\mathbf{E} = \frac{x}{x^2 + y^2}\mathbf{a}_x + \frac{y}{x^2 + y^2}\mathbf{a}_y$$

Then $\frac{dy}{dx} = \frac{E_y}{E_x} = \frac{y}{x}$ or $\frac{dy}{y} = \frac{dx}{x}$

whose solution is $\ln y = \ln x + C_1$ or $\ln y = \ln x + \ln C$

Finally $y = Cx$



3: Electric Field Intensity 电场强度



- Streamlines and sketches of fields 电力线和电场分布图

Methodology of Streamline Construction 电力线的构建方法

D2.7. Find the equation of that streamline that passes through the point $P(1, 4, -2)$ in the field $\mathbf{E} = (a) \frac{-8x}{y} \mathbf{a}_x + \frac{4x^2}{y^2} \mathbf{a}_y$; (b) $2e^{5x} [y(5x + 1) \mathbf{a}_x + x \mathbf{a}_y]$.

Ans. $x^2 + 2y^2 = 33$; $y^2 = 15.7 + 0.4x - 0.08 \ln(5x + 1)$

3: Electric Field Intensity 电场强度



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$$(a) \quad E_x = \frac{-8x}{y} \quad E_y = \frac{4x^2}{y^2} \quad \frac{dy}{dx} = \frac{E_y}{E_x} = -\frac{4x^2}{y^2} \frac{y}{8x} = -\frac{x}{2y}$$

$$2ydy = -xdx \quad y^2 = -\frac{1}{2}x^2 + C \quad \text{or} \quad 2y^2 + x^2 = C'$$

For $P(1,4,-2)$, $C'=33$

HW-2



2. Three point charges are positioned in the x - y plane as follows: 5 nC at $y = 5\text{ cm}$, -10 nC at $y = -5\text{ cm}$, 15 nC at $x = -5\text{ cm}$. Find the required x - y coordinates of a 20-nC fourth charge that will produce a zero electric field at the origin.
3. Eight identical point charges of $Q\text{ C}$ each are located at the corners of a cube of side length a , with one charge at the origin, and with the three nearest charges at $(a, 0, 0)$, $(0, a, 0)$, and $(0, 0, a)$. Find an expression for the total vector force on the charge at $P(a, a, a)$, assuming free space.
4. A 100 nC point charge is located at $A(-1, 1, 3)$ in free space. a) Find the **locus** of all points $P(x, y, z)$ at which $E_x = 500\text{ V/m}$; b) Find y_1 if $P(-2, y_1, 3)$ lies on that locus. (**Locus**: 轨迹面, 请用面的方程来表示)
5. A uniform volume charge density of $0.2\text{ }\mu\text{C/m}^3$ is present throughout the spherical shell extending from $r = 3\text{ cm}$ to $r = 5\text{ cm}$. If $\rho_v = 0$ elsewhere: a) find the total charge present throughout the shell; b) find r_1 if half the total charge is located in the region $3\text{ cm} < r < r_1$.